

RAM KRISHNA YADAV

FULL STACK SOFTWARE DEVELOPER

[Linkedin](#) | [Portfolio](#) | itsramky234@gmail.com | [Github](#)

SUMMARY

Full Stack Software Developer who builds production-grade web systems with TypeScript, Node.js, and PostgreSQL. Creator of **Zyotra**, an agentless automated deployment PaaS, and author of an open-source **JavaScript thread-pool library** with **10,000+ npm downloads**. Comfortable designing scalable REST APIs, **role-based systems**, and **WebSocket real-time architectures**, with hands-on experience building and deploying full-stack applications from development to production.

SKILLS

- Languages : Javascript, Typescript, Python, C++
- Frontend : React, Redux Toolkit, Tailwind CSS, HTML5
- Backend : Node.js, Express.js, Elysia.js, REST APIs, JWT, **WebSockets**
- Databases : PostgreSQL, MongoDB, MySQL, ORMs, **Redis**, **Pub/Sub**
- AI / ML : **RAG Pipelines**, Vector Embeddings, pgvector, Semantic Search, LLM Integration, MCP, Vercel AI SDK
- Systems and Tools : **Worker Threads**, Event-Driven Architecture, **Docker**, **AWS**, **VPS Deployment**, Git, Jest Testing

PROFESSIONAL EXPERIENCE

Software Engineer at Venyx Technology International | Mar 2026 - Present

- Built and shipped management software for restaurants and small businesses, contributing across the stack with React.js, Node.js, and TypeScript
- Designed and implemented **REST APIs** backed by **PostgreSQL**, writing schemas and queries for reliable, structured data handling
- Implemented **JWT-based authentication** and role-based access to secure application endpoints and user data
- Deployed and maintained applications on **VPS and AWS EC2** environments, handling configuration and production releases
- Collaborated within a development team to translate requirements into working features and iterate based on feedback

PROJECTS

EliteNotes — AI Sales Assistant Platform with RAG Pipeline [Live](#)

Tech : Next.js, React, TypeScript, PostgreSQL, Drizzle ORM, pgvector, Redis, Vercel AI SDK, Cloudflare R2, Docker

- Built a production-grade AI sales assistant platform unifying CRM data and unstructured documents through a **RAG pipeline**, persistent memory, and intelligent tool orchestration
- Engineered a multi-format document ingestion pipeline (PDF, DOCX, XLSX, CSV) with recursive text chunking and **768-dimension Gemini vector embeddings** stored in pgvector for semantic search
- Implemented vector similarity search with configurable top-K retrieval, similarity thresholds, and source-cited AI responses backed by retrieved document context
- Integrated **120+ third-party services** (Salesforce, HubSpot, Google Drive, Slack) via Composio and the Model Context Protocol (MCP) with unified OAuth and dynamic tool loading
- Developed multi-provider AI chat with **real-time SSE streaming**, multiple auth strategies (Google OAuth, Magic Link), token/cost telemetry, and Redis caching for session performance

Zyotra — Self-Hosted PaaS & Automated Deployment Platform [Live](#) | [Github](#)

Tech : TypeScript, Bun, Node.js, Nginx, PostgreSQL, MySQL, Redis, WebSockets, AWS S3, Docker

- Built a self-hosted PaaS replicating the Vercel/Heroku experience on bare-metal servers, cutting cloud infrastructure costs by up to **80%**
- Engineered **agentless deployment automation** via SSH scripts handling Git cloning, SSL termination (Certbot), and **Nginx reverse proxy configuration**
- Architected **real-time build log streaming** and live **machine analytics** (CPU/RAM) over WebSockets for deployment observability
- Built a **Database Management System** with automated PostgreSQL/MySQL backups to AWS S3, one-click Redis deployment, and an integrated **SQL Query Runner**

ChessOnline - Realtime Multiplayer Chess Game [Live](#) | [Github](#)

Tech : React.js, Node.js, Express.js, WebSocket, MongoDB, Tailwind CSS, TypeScript

- Built a real-time multiplayer chess application with **low-latency WebSocket communication**
- Implemented **private rooms**, turn-based gameplay, and authoritative server-side game-state consistency
- Developed a responsive chessboard UI with move validation and complete chess-rule enforcement
- Designed a scalable backend to **manage concurrent game sessions** and **real-time player matching**
- Engineered an event-driven socket layer to broadcast moves, sync game state, and handle player disconnects/reconnects gracefully

Thready.js — Open-source JavaScript Thread Pool Library [Live](#) | [Github](#)

Tech : TypeScript, NPM Package

- Designed a type-safe **thread-pool abstraction** for Web Workers and Node.js worker_threads
- Implemented **worker pooling**, **task queueing**, and a **zero-dependency** architecture
- Abstracted low-level concurrency patterns into a developer-friendly API, demonstrating strong understanding of JavaScript's **event loop and worker model**
- Achieved **10,000+ npm downloads**, indicating real-world adoption

EDUCATION

Bachelor of Computer Engineering (2024–2028)

Tribhuvan University, IOE Purwanchal Campus, Nepal